



Anonymous Connections and Onion Routing

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Subjects: Radio Communications ; *SECURE COMMUNICATIONS ; DATA PROCESSING ; WIDE AREA NETWORKS ; ROUTING ; ANONYMOUS ; *ONION ROUTING

DDC: 629 Other branches of engineering (computed)

Description

Onion Routing is an infrastructure for private communication over a public network. It provides anonymous connections that are strongly resistant to both eavesdropping and traffic analysis. Onion routing's anonymous connections are bidirectional and near real time, and can be used anywhere a socket connection can be used. Any identifying information must be in the data stream carried over an anonymous connection.

An onion is a data structure that is treated as the destination address by onion routers; thus, it is used to establish an anonymous connection. Onions themselves appear differently to each onion router as well as to network observers. The same goes for data carried over the connections they establish. Proxy aware applications, such as web browsing and e-mail, require no modification to use onion routing, and do so through a series of proxies.

A prototype onion routing network is running between our lab and other sites. This paper describes anonymous connections and their implementation using onion routing. This paper also describes several application proxies for onion routing, as well as configurations of onion routing networks.

Publication Details

- **Contributors:** NAVAL RESEARCH LAB WASHINGTON DC CENTER FOR HIGH ASSURANCE COMPUTING SYSTEMS (CHACS)
- **Year of Publication:** 1998
- **Document Type:** Text ; [Text]
- **Source:** DTIC
- **Language:** en
- **Rights:** Approved for public release; distribution is unlimited.
- **Content Provider:** Defense Technical Information Center: DTIC Technical Reports database

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